

“AI-Based Smart Traffic Light System”

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Outline

1. Introduction
2. Problem Statement & Motivation
3. Objectives
4. Literature Survey
5. Existing vs. Proposed System
6. Methodology & Architecture
7. Result & Conclusion

INTRODUCTION

Overview: The proposed system is an AI-based smart traffic light system that monitors traffic using a camera and automatically adjusts signal timing according to traffic conditions.

Current Scenario: Currently, traffic signals operate on fixed timing systems. These systems do not consider real-time traffic density, leading to traffic congestion, longer waiting times, and fuel wastage.

The Concept: The system detects vehicles using YOLO, calculates traffic density for four lanes, and uses Reinforcement Learning to select suitable signal actions.

PROBLEM STATEMENT & MOTIVATION

Problem Statement: Currently, traffic light systems suffer from inefficiency due to fixed timing control, lack of real-time traffic analysis, and absence of emergency vehicle prioritization, leading to congestion and delays.

Motivation:

- Reduce waiting time.
- Improve traffic flow.
- Provide adaptive signal control.
- Give priority to emergency vehicles.

OBJECTIVES

- Detect vehicles using YOLO.
- Count vehicles in 4 lanes.
- Calculate traffic density.
- Use Reinforcement Learning for signal decisions.
- Generate dynamic signal timing.
- Control traffic lights using ESP32.
- Provide emergency vehicle priority.
- Monitor traffic data using Firebase.

LITERATURE SURVEY

Ref	Paper Title / Year	Key Methodology	Limitation / Gap
[1]	"Adaptive Traffic Signal Control System Using Object Detection Based Approach (2025)"	YOLOv8 for vehicle detection; vehicle counting; adaptive signal control; emergency vehicle priority	Mainly focuses on object detection and signal adjustment; limited to a single intersection
[2]	"Dynamic Traffic Control System Using YOLO-Based Model Considering Vehicle Count and Weather Conditions (2025)"	YOLO-based vehicle detection and counting; dynamic signal timing; weather conditions	Depends on external weather data; does not use Reinforcement Learning for timing optimization
[3]	"MetaSignal: Meta Reinforcement Learning for Traffic Signal Control via Fourier Basis Approximation (2025)"	"Reinforcement Learning; adaptive signal control; learning from different traffic scenarios"	Requires training and adaptation; does not include vehicle detection or hardware implementation

Research gap/Existing Limitations

- Fixed-time traffic signals.
- Limited real-time vehicle detection.
- No lane-wise traffic analysis.
- Limited adaptive signal control.
- Emergency priority is not always available.
- Limited integration of AI, IoT and cloud monitoring.

PROPOSED METHODOLOGY

Approach: Traffic video is given as input to the YOLO model. Detected vehicles are counted for four lanes and the traffic density is calculated.

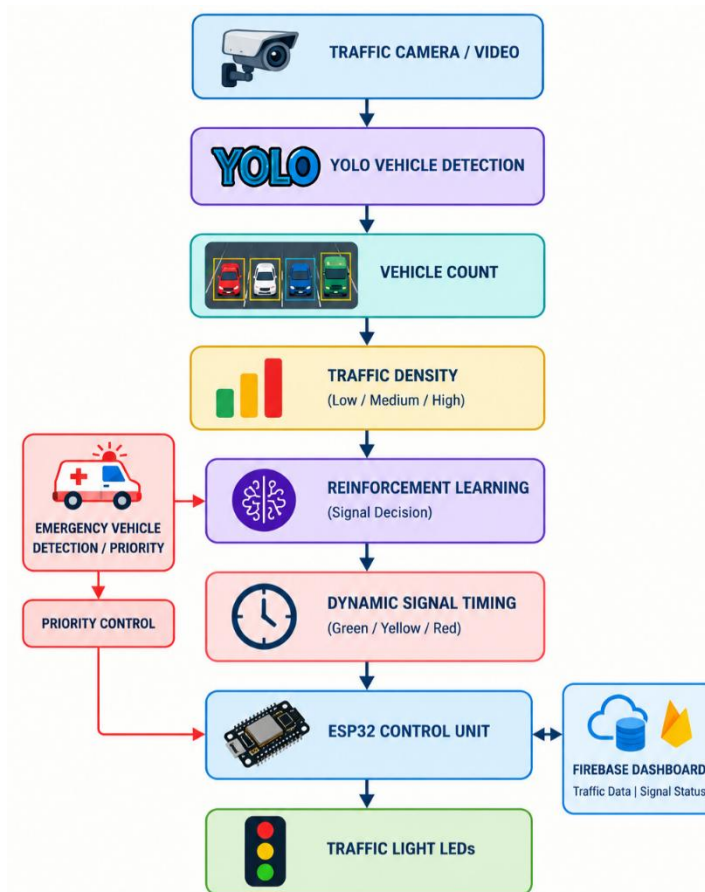
Hardware Stack:

- ESP32 Microcontroller
- Camera Module
- Traffic LEDs
- Push Button (Emergency)
- Breadboard & Jumper Wires

Software Stack:

- Arduino IDE
- Embedded C
- Python (AI Model)
- YOLO(ulalytics)
- Firebase (Cloud Platform)
- Libraries: NumPy, TensorFlow

SYSTEM ARCHITECTURE



WORK CARRIED OUT & FUTURE PLAN

Completed in Phase-I:

- Literature Survey & Requirement Analysis.
- Feasibility Study & Tool Selection.
- High-level System Design.

Planned for Phase-II:

- Hardware setup and coding
- AI model implementation
- Integration of modules
- Testing and result analysis

Technology Stack

- Programming: Python, C/C++
- AI & Computer Vision: YOLO, OpenCV
- Machine Learning: Reinforcement Learning
- Hardware: ESP32, Traffic Light LEDs, Camera Module
- Database & Dashboard: Firebase
- Development Tools: Arduino IDE, VS Code

Implementation – Part 1

Vehicle Detection:

- Traffic video used as input.
- YOLO detects vehicles.
- Bounding boxes are generated.
- Vehicle classes are identified.

Vehicle Counting:

- Cars, Motorcycles, Buses, Trucks

Lane Analysis:

- Lane 1, Lane 2, Lane 3, Lane 4

Traffic Density: Vehicle counts are converted into traffic-density. **Levels: LOW → MEDIUM → HIGH**

The calculated density represents the current traffic condition of each lane.

RESULTS

Initial Results:

- YOLO successfully detects vehicles.
- Vehicle count is obtained from traffic video.
- Traffic density is calculated. 4-lane structure is defined.
- Signal timing can be adapted according to traffic density.

Output:

- Vehicle detection.
- Lane-wise count.
- Traffic density.
- Signal decision.

Implementation – Part 2

- Reinforcement Learning
- Traffic state as input.
- Signal phase as action.
- Reward based on traffic flow.
- Learns suitable signal decisions.
- Dynamic Signal Control.
- Green timing changes according to traffic.
- Yellow and red phases are controlled.
- Lane-wise signal management.

Implementation – Part 2

- Hardware & IoT
- ESP32 controls signal LEDs.
- Emergency input provides priority.
- Firebase stores traffic information.
- Dashboard displays signal status and traffic data.

REFERENCES

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THANK YOU